FINDING MEMO

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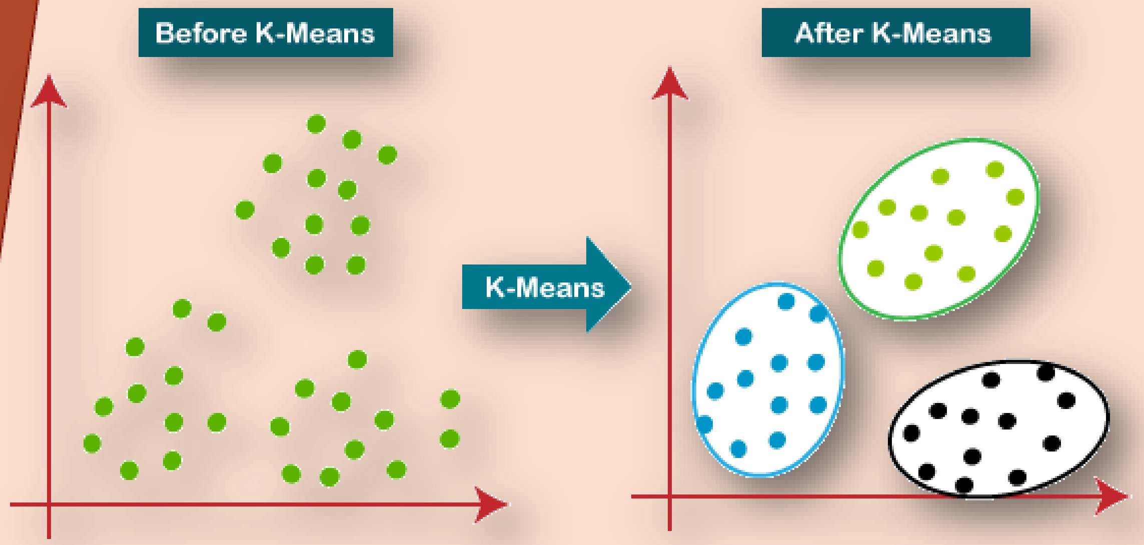
K-means Clustering



Definition

- **Unsupervised ML algorithm** that partitions dataset into *K distinct clusters* based on **feature similarity**

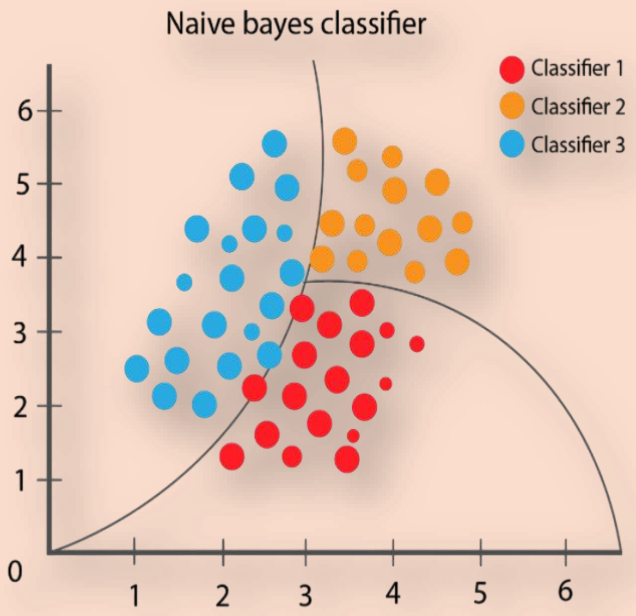
How It Works (Basic Steps)

1. Choose K and Initialize Centroids
2. Assign data points to centroids
3. Update centroids
4. Convergence Check
5. Final output

Main idea

- Minimize the total within-cluster **sum of squared distances** between each data point and the **mean** of its cluster

Naive Bayes

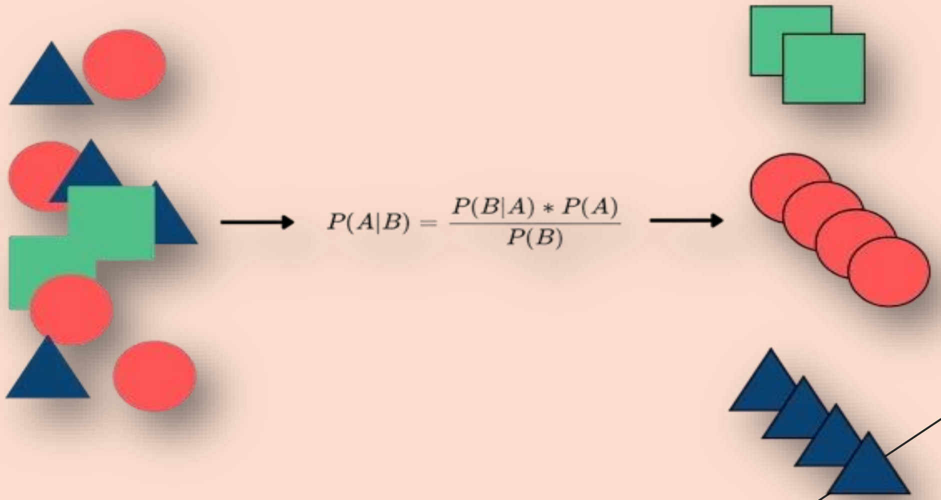


Definition & Main Idea

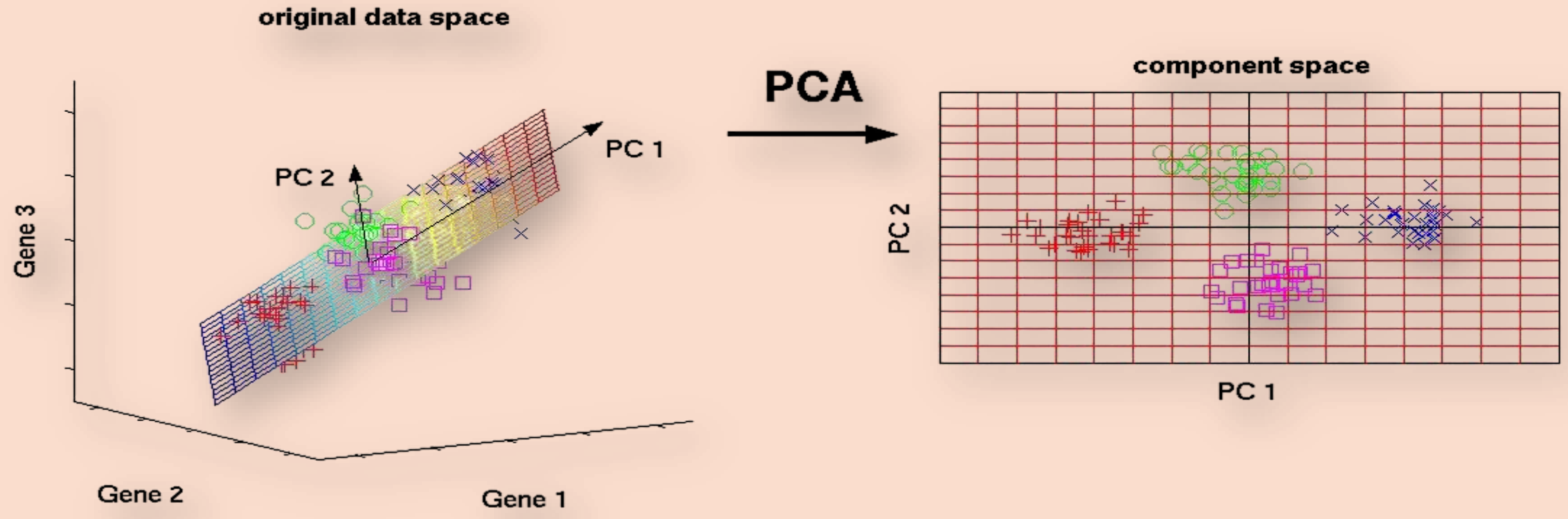
- A simple, fast probabilistic classifier that based on Bayes' theorem under the assumption that features are independent given the class.
- Despite this "naïve" assumption, it performs very well on **high-dimensional** tasks like classification.

How it works (basic steps)

1. Calculate Priors
2. Estimate Likelihoods
3. Classify new data



Principal Component Analysus (PCA)

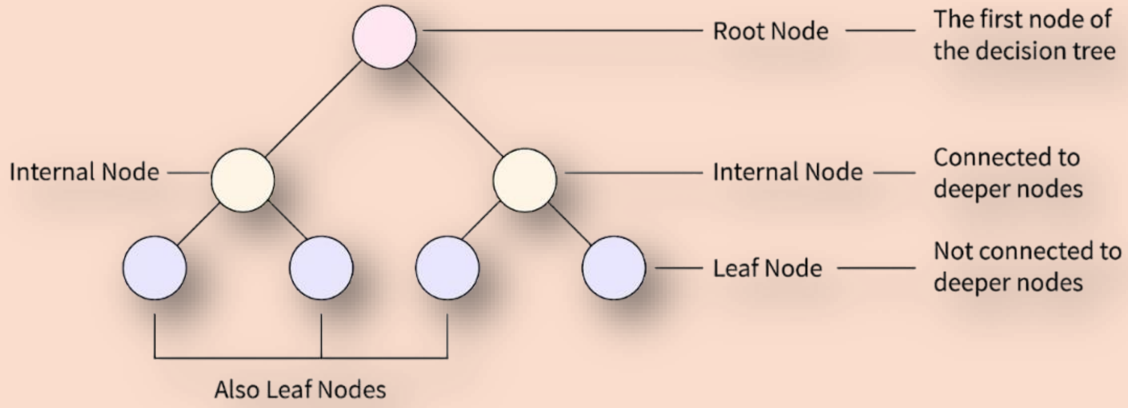


- **Definition:** a technique that transforms a dataset into a new set of uncorrelated variables, ordered by the amount of original variance they capture.
- **Main Idea:** PCA finds the directions that explain the most variance, allowing you to compress or visualize high-dimensional data with minimal information loss.

How it works (basic steps)

1. Standardize the Data
2. Compute the Covariance Matrix
3. Eigen-Decomposition
4. Select Top *k* Components
5. Project Data

Decision Tree



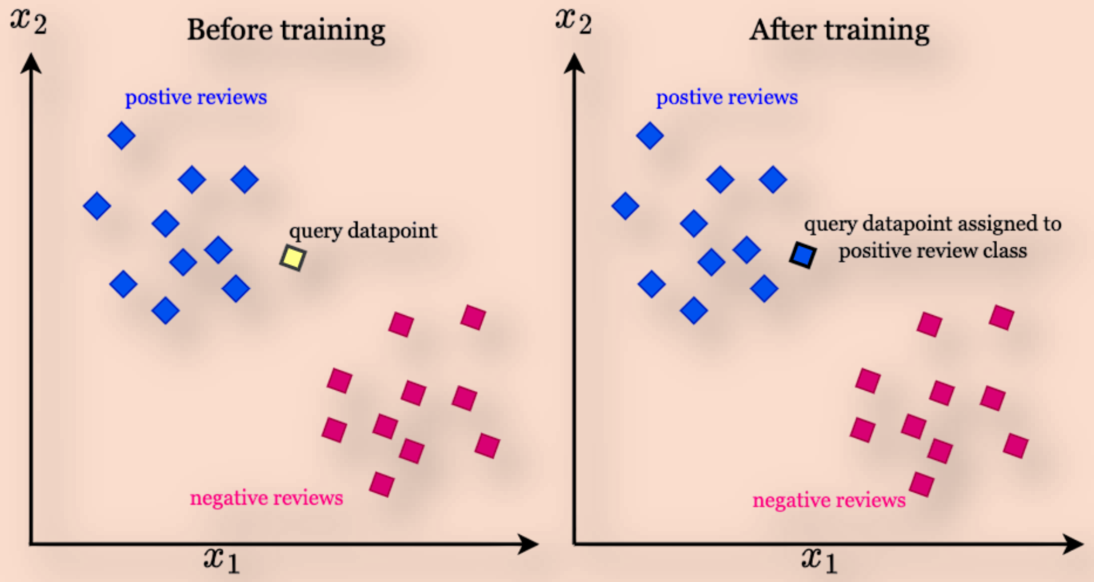
Definition & Main idea

- A decision tree is a supervised learning model that splits data via feature-based tests at each node to form a tree whose leaves yield class labels or continuous predictions.

How it works (basic steps)

1. Start with the full dataset at the root
2. Select the best feature and threshold
3. Partition the data
4. Repeat recursively
5. Assign leaf predictions

k-Nearest Neighbours

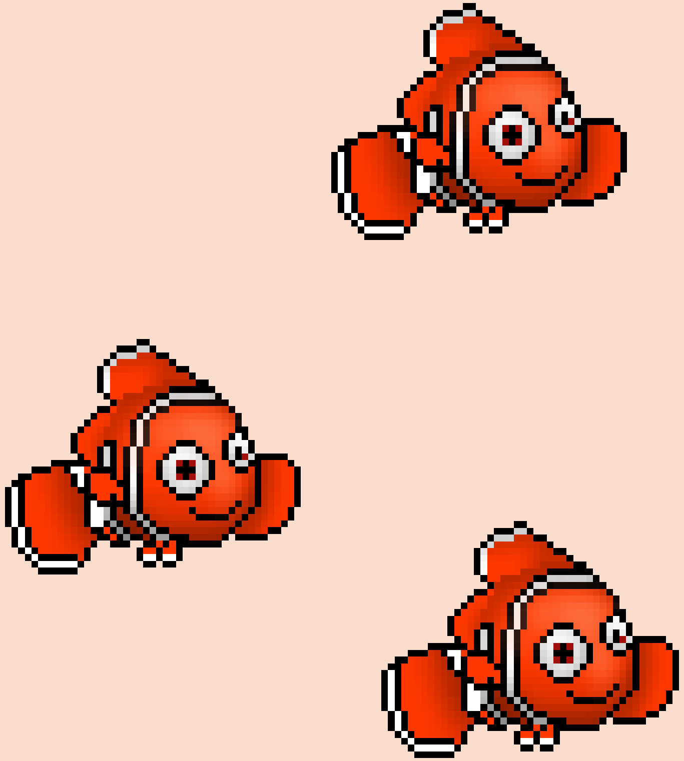


- **Definition & Main idea**

A non-parametric, instance-based method that assigns a label or value to a new point by looking at the labels/values of its k closest training examples in feature space, on the assumption that nearby points are alike

How it works (basic steps)

1. Choose k
2. Compute Distances
3. Find k Nearest Neighbors
4. Aggregate their Labels/Values
5. Return Prediction



Thank you for your attention!